

Latch-ups (SEL). Additionally, headers were added for extra flexibility in digital and analog monitoring capabilities. All the components utilized in the design were selected and tested to the simulated radiation levels for the HL-LHC lifetime plus safety factors. The single point of failure mitigation strategies to reduce the impact radiation effects include: redundancy layers, GBTx ASICs, Triple Mode Redundancy (TMR), the Xilinx Soft Error Management (SEM), and GBT-FEC protocol.

III. THE SIMULATED RADIATION ENVIRONMENT

The ATLAS Radiation Tolerant Electronics Proposed Guideline on COTS Lot to Lot Variation [4] was put together by the collaboration as a policy for better consistency in the radiation qualification tasks within the different sub-detector electronics designs [4]. Due to the cylindrical symmetry of the ATLAS detector the radiation requirements were extracted from the radiation levels of the DBs positioned in the most heavily irradiated environments [5]. The minimum Total Ionizing Dose required for qualification of each of the design components is 108 Gy of, with inclusion of safety factors of 1.5 for simulation and 3.0 for lot variation in the components. The Non-Ionizing Energy Losses (NIEL) fluence required for qualification is 13.16×10^{12} n/cm², taking into account safety factors of 1.5 for simulation, 1.3 when using a non-monoenergetic neutron source, and 3.0 to account for lot variation of the components. The total expected fluence for Single Event Effects (SEE) is 4.54×10^{11} h/cm², where a safety factor of 3.0 for simulation was included in the calculations.

IV. DAUGHTERBOARD RADIATION QUALIFICATION STUDIES

A diverse set of radiation campaigns took place were commercial development boards, dedicated boards, and full four assembled DB prototypes that were irradiated to more than 108 Gy and 14×10^{12} n/cm². These tests qualified the Kintex Ultrascale FPGAs, the ProASIC FPGAs, the IS25LP128F, IS25WP256 and IS25LP512 flash memories, DAN217T146CT Schottky diodes, 540BAA100M000BBG oscillators, and INA333AIDRG instrumentation amplifiers. The ProASIC FPGAs and IS25LP128F flash memories showed some after-effects on the re-programmability of the devices that were fully recovered in all devices after annealing and have no impact in the nominal run of the system. Similarly, the impact of ionizing radiation on the threshold voltages the FDFMA3N109 MOSFETs is acceptable for the digital use of these devices on the design. In the case of the QUAD oscillators LTC6902IMS it was determined that the inconsistent post irradiation behavior over the sample set was sufficient for disqualifying the device. Four different variants of SFP+ devices from four different companies were tested: AVAGO, CORETEK, FS-Europe and Direktronik. The AVAGO AFBR-709SMZ the CORETEK CT-A000NPP-SB1L-D devices failed to pass the irradiation tests, while both the Direktronik 33-4248 and the FS-Europe SFP-10GSR-85 were qualified with

some of the samples being able to reach 21×10^{12} n/cm² and more than 200 Gy without any failures.

For SEEs three different tests were performed for studying the SELs and Single Event Upsets (SEUs). A KU+ FPGA was irradiated to 2.2×10^{11} p/cm² with 54 MeV protons, where a SEU rate of 24 SEU/(10⁹ p/cm²) and an SEL rate of 2 SEL/(10⁹ p/cm²). The presence of SEL in the KU+ FPGA disqualified it for usage on any on-detector design. In the case of the KU FPGA two tests were done with 54 MeV protons and 226 MeV protons where no SEL were observed up to 2.62×10^{11} p/cm² and 3.2×10^{11} p/cm² respectively. As for the SEU rates, we measured 116 SEU/(10⁹ p/cm²) and 80 SEU/(10⁹ p/cm²). Worst case scenario, the SEU rates represent roughly 5498 SEUs per year out of which approximately 5487 are correctable by the SEM and won't impact the nominal runs due to the use of TMR. The remaining 11 uncorrectable errors per year will be managed by the TMR and can be removed by reloading the FPGA configuration from the FLASH memory during data acquisition stops.

V. CONCLUSIONS

The radiation tests performed have qualified the key components of the DB design for TID and NIEL resistance up to the required radiation levels. Additionally, SEE tests have allowed to eliminate the possibility of SEL on the KU FPGA used in the current revision of the design and estimate the impact of SEUs on the data taking for nominal runs. The team plans conducting radiation tests in mixed fields and additional SEU tests with operational firmware to further evaluate the impact of SEU rates on board performance. Additionally, further radiation tests will be required during production phase to account for variance in the batches during the production of components. The tests will follow the ATLAS Radiation Tolerant Electronics Proposed Guideline. The project will produce 930 DBs of which 896 will be installed in ATLAS for the HL-LHC.

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